

Youssef Awad

youssef-awad@outlook.com | +971507399008 | UAE (GMT+4)

linkedin.com/in/youssef-awad | github.com/DeadPackets | deadpackets.pw | projects.deadpackets.pw

Summary

Software engineer whose work spans the full stack, DevOps, and security, with a strong focus on agentic AI and an offensive-security background (OSCP). I co-founded CTFae and built the infrastructure behind the world's largest CTF: 10,000+ concurrent users, a Guinness World Record, and Best Emirati Startup at GISEC 2024. I solve real problems with code, move between cloud infrastructure, backend, frontend, and exploit development, automate the boring parts, and threat-model everything I build like an attacker.

Core strengths: full-stack engineering and platforms at scale, offensive security (OSCP), agentic AI systems, and solving hard problems end to end.

Professional Experience

Head of Engineering & Co-Founder | CTFae – Dubai, UAE 2021 – Present

- Architected the Challenge Deployment API (Python/FastAPI) for on-demand provisioning of containerized challenges to AWS ECS with auto-scaling, serving 10,000+ concurrent participants at BlackHat MEA (Guinness World Record, 2024).
- Built the CyberRanges IaC platform (Terraform, Packer, Ansible) for automated multi-VM attack-defense environments across AWS and GCP, with GitHub Actions CI/CD and OIDC-based cloud authentication.
- Developed an attack-defense checksystem (Python, Celery, PostgreSQL with PgCat pooling, Redis) for real-time scoring of 100+ teams simultaneously.
- Owned full-stack platforms (FastAPI, Next.js) end to end: infrastructure, backend, frontend, and observability, all kept reliable under live event load.
- Led engineering for 30+ global events, including the first Bug Bounty CTF at DEFCON 33, GITEX Dubai, the Bahrain National CTF, BSides Nashville (USA), HITB, and Dubai Police CTF.

Senior Cyber Security Consultant | Malcrove – Dubai, UAE 2022 – Dec 2025

- Led penetration testing for UAE government and critical-infrastructure clients across web/mobile applications, internal/external networks, and wireless systems.
- Achieved a 90%+ success rate in obtaining Domain Admin privileges across 100+ Active Directory engagements.
- Authored CVSS-scored vulnerability reports with proof-of-concept exploits and remediation guidance, and briefed non-technical executive leadership on risk.

Volunteer | GDI.Foundation – Remote 2019 – 2024

- Contributed to responsible-disclosure and large-scale internet vulnerability remediation efforts for a non-profit focused on internet security.

Selected Projects

Agentic AI & LLM Systems

- **ProjectHailMary** – Solo-operated agentic vulnerability finder: one orchestrator drives 60+ charter-bearing subagents through a validation gauntlet (trace, PoC, CVSS, dedup) into a deterministic CLEAR/HOLD/REJECT verdict. 23 modules, 465+ tests. (Python)
- **LLM-AD-Defense** – Autonomous SOC analyst on Gemini 2.5 Pro: ingests Wazuh EDR alerts, reasons about the threat, and executes its own remediation, down to isolating a compromised Active Directory account. (Python/FastAPI)
- **PiCodeReview** – Multi-tier agentic code reviewer: cheap-model specialist hunters over-report, then a frontier-model judge re-reads the code and prunes the false positives. Built as a 30-minute take-home. (TypeScript)
- **LLMChessArena** (live) – Any LLM vs any LLM, Stockfish, or a human over OpenRouter, with real-time WebSocket streaming, move-by-move Stockfish evaluation, and per-move token/cost tracking. (TypeScript/React)
- **BuzzerMinds** (live) – Live multiplayer trivia game where every question is LLM-generated and grounded against real sources via SearXNG, with fact cards and strict schemas to keep the AI honest. (Next.js/FastAPI)

Platform, Infrastructure & Security Tooling

- **Challenge Deployment API, CyberRanges, Attack-Defense Checksystem** – production CTFae platforms operating at 10,000+ user scale (see Experience).
- **pwnbox** – Kali-based Docker container pre-loaded with offensive tooling, ZSH, and SSH for instant, reproducible pentest environments. (Shell/Docker)
- **InfoSteal, HistoryThief, EmailCheck** – browser-based security demos showing client-side information leakage, browser-history exfiltration via JS/CSS tricks, and multi-API breach checking. (JavaScript)

Data, Web & Open Source

- **VisualizeAUS** (live) – 20 years of university course data turned into an interactive exploration. (Python, Plotly, Quarto)
- **AUSCrawl** – crawler that pulls a Banner registration system into a clean, analyzable dataset, and a solid data-science playground. (Python)
- **DOSArchive** (live) – preserved my father's old DOS programs, emulated to run directly in the browser. (Astro)

Technical Skills

Applied AI / LLMs: Multi-agent orchestration, RAG, evals & guardrails, prompt engineering; Claude, Gemini, Codex, Open-Router

Languages: Python, TypeScript, JavaScript, Bash, SQL

Backend & APIs: FastAPI, Hono, Flask, Express, PostgreSQL, Redis, RabbitMQ, Celery, Pydantic, SQLAlchemy, Nginx

Frontend: React, Next.js, Vite, Tailwind

Cloud & DevOps: Docker, Terraform, Ansible, Packer, GitHub Actions, AWS (ECS, ECR, IAM), GCP, Cloudflare, Linux

Security: Penetration testing, red teaming, Active Directory, vulnerability assessment, exploit development, CTF design

Education & Certifications

B.Sc. Computer Engineering (Minor: Computer Science) – American University of Sharjah, UAE

2017 – 2022

OSCP – Offensive Security Certified Professional (Score: 90/100)

Speaking, Training & Community

- **HITB SecConf (Thailand, 2024):** delivered CTFae's first Cyber Drill workshop.
- **CPX CyberAlley (2024):** invited talk on CTFae's story, growth, and engineering.
- **HITB SECCONF (2021):** ran a two-day hands-on training on building Attack-Defense CTFs.
- **GISEC (2020):** live demonstration of a Windows credential-stealing attack.
- Cybersecurity educator and community builder; created the CrackTheHack educational series of interactive security demos.

Awards & Recognition

- **Guinness World Record (2024)** – largest CTF competition in the world, BlackHat MEA.
- **Best Community Engagement (2025)** – UAE CyberSecurity Awards.
- **Best Emirati Startup (2024)** – GISEC, UAE • **Official CTF Partner for GISEC UAE (2024).**
- **Outstanding CSE Student Talent Award (2021)** – American University of Sharjah.

Competition Wins & Events Engineered

- **Events engineered and hosted:** the first Bug Bounty CTF at DEFCON 33 (2025), the Dubai Cyber Security Challenge at GITEX (2025), the Bahrain National CTF (2025), BSides Nashville (2024–2025, first USA CTFs), BlackHat Saudi Arabia, HITB Amsterdam, GITEX Africa, the Dubai Police CTF, and the largest CTF in the Middle East at AtHack.
- **First-place wins:** The Hunt SOCathon @ GISEC (2021), CyberKnight CTF (2020), CyberTalents UAE Regional (2019–2020), HITB Lockdown (2020), CyberBattle of the Emirates (2019), Dubai Police CTF (2019), and CyberQuest Attack & Defence (2016, 2018).
- **Additional placements:** 2nd, HITB AI Challenge (2019); 2nd, NYU CSAW Forensics (2016); 3rd, CyberTalents Arab Regional (2019).